HOMEWORK 4
NUMERICAL METHODS
FALL 2025

DUE FRIDAY, OCT 17

Instructions: Consider the following problems.

Prob 1. An interpolating polynomial of degree 20 is to be used to approximate e^{-x} on the interval $[0, 2]$. How accurate will it be ?

Computer Prob 1. Consider the **Runge** function given by

$$f(x) = \frac{1}{(1 + x^2)}$$

perform the following tasks:

a) Construct and plot a polynomial of degree 10, that is $P_{10}(x)$ using equally spaced points on the interval $[-5, 5]$.

b) Same task as part a) but now consider ChebyShev nodes and label your polynomial as $G_{10}(x)$.

c) Plot $f(x)$, $P_{10}(x)$ and $G_{10}(x)$ vs x in one figure and clearly label your functions. You should use a fine mesh, that is, use more than 11 points so that your plots are smooth.

d) To test things out, plot $G_{20}(x)$, $G_{40}(x)$ and $G_{60}(x)$ and discuss your findings. Provide an educated guess of your results. Fix your window axis([-5 5 -0.5 2])

Note: The above tasks are to be done with the codes developed in class: *Coef and Eval.*

Note1: The programming should be done in terms of a Script and a function file in Matlab. Your submission should include a folder named "YourLastName" that contains 1) a .txt file with basic instructions/description of your script.m and your function.m file 2) script.m file and 3) function.m file. It is important to compress your folder before submitting it on Canvas. Try something like Winzip or Winrar. **In addition to the compressed folder with all content you also need to submit a .pdf file which contains your homework problems in project format.** **Note2:** Show all your work. **Note3:** Yes, we will be running your codes.
Use graphical representation whenever possible to support summaries.

*****Please follow instructions carefully - submit on time/no late acceptance *****
